

DATA CLEANING

```
In [16]: #step1: Load dataset
import pandas as pd
df = pd.read_excel("Data_Cleaning_50Plus_Records.xlsx")
print(df.head())

#2 Renaming Columns
#Check current Column Names
print(df.columns)

df.rename(columns={
    'Student ID': 'Student_ID',
    'Student Name ': 'Student_Name',
    ' Department': 'Department'
}, inplace=True)
print(df.columns)

#3.Changing Data Types
#Check Data Type
print(df.dtypes)

#Convert Student_ID to String
df['Student_ID'] = df['Student_ID'].astype(str)

#Convert Age to Integer
#First handle Missing values
df['Age'] = df['Age'].fillna(0)
df['Age'] = df['Age'].astype(int)

#Check
print(df.dtypes)
```

	Student ID	Student Name	Department	Age	Salary	Email
0	101	student1	IT	20.0	23075.0	student1@gmail.com
1	102	student2	ECE	18.0	35929.0	student2@gmail.com
2	103	STUDENT3	EEE	22.0	29718.0	student3@gmail.com
3	104	student4	MECH	20.0	21009.0	student4@gmail.com
4	105	Student5	CSE	19.0	47113.0	student5@gmail.com

```

Index(['Student ID', 'Student Name ', ' Department', 'Age', 'Salary', 'Email'], dtype='object')
Index(['Student_ID', 'Student_Name', 'Department', 'Age', 'Salary', 'Email'], dtype='object')
Student_ID      int64
Student_Name    object
Department      object
Age             float64
Salary          float64
Email           object
dtype: object
Student_ID      object
Student_Name    object
Department      object
Age             int64
Salary          float64
Email           object
dtype: object

```

```

In [15]: #4.Detecting Duplicate Records
#Find Duplicates
duplicates = df.duplicated()
print(duplicates)

#Count Duplicates
print(df.duplicated().sum())

#5.Drop Duplicates Records
#Before
print(df.shape)

#Remove Duplicates
df.drop_duplicates(inplace=True)

#After
print(df.shape)

```

0	False
1	False
2	False
3	False
4	False
5	False
6	False
7	False
8	False
9	False
10	False
11	False
12	False
13	False
14	False
15	False
16	False
17	False
18	False
19	False
20	False
21	False
22	False
23	False
24	False
25	False
26	False
27	False
28	False
29	False
30	False
31	False
32	False
33	False
34	False
35	False
36	False
37	False
38	False
39	False
40	False
41	False

```
42    False
43    False
44    False
45    False
46    False
47    False
48    False
49    False
50    False
51    False
dtype: bool
0
(52, 6)
(52, 6)
```

```
In [18]: #6.String Operations
         #string methods work only on text columns.
         #Remove extra spaces
         df['Student_Name'] = df['Student_Name'].str.strip()

         #convert to uppercase
         df['Student_Name']=df['Student_Name'].str.upper()
         print(df['Student_Name'].head())

         #convert to TitleCase
         df['Student_Name']=df['Student_Name'].str.title()
         print(df['Student_Name'].head())
```

```
0    STUDENT1
1    STUDENT2
2    STUDENT3
3    STUDENT4
4    STUDENT5
Name: Student_Name, dtype: object
0    Student1
1    Student2
2    Student3
3    Student4
4    Student5
Name: Student_Name, dtype: object
```

In [36]: `print(df.columns)`

```
#Replace Text
#Replace department name.
df[' Department']=df[' Department'].str.replace(
    'CSE',
    'Computer Science'
)
print(df['Department'].head())

#Replace Multiple Values - No Need str
df[' Department']=df[' Department'].replace({
    'CSE':'Computer Science',
    'EEE':'Electronics',
    'Information Technology':'IT',
    'MECH':'Mechanical'
})
print(df[' Department'].head())
```

```
Index(['Student_ID', 'Student_Name', 'Department', 'Age', 'Salary', 'Email',
      ' Department'],
      dtype='object')
0    Information Technology
1           Electronics
2                EEE
3                MECH
4        Computer Science
Name: Department, dtype: object
0                IT
1        Electronics
2        Electronics
3        Mechanical
4    Computer Science
Name: Department, dtype: object
```

In [39]:

```
#String Length
df['Name_length']=df['Student_Name'].str.len()
print(df[['Student_Name', 'Name_length']].head())
```

	Student_Name	Name_length
0	Student1	8
1	Student2	8
2	Student3	8
3	Student4	8
4	Student5	8

```
In [45]: #Extract First Character
df['First_letter']=df['Student_Name'].str[0]
print(df[['Student_Name','First_letter']].head())
```

	Student_Name	First_letter
0	Student1	S
1	Student2	S
2	Student3	S
3	Student4	S
4	Student5	S